Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Entropy is not continuous in standard probability spaces

Files.

- File: `briefs/EntropyTopology/EntropyTopology.lean`
- Library: `EntropyTopology`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, `Mathlib.Analysis.SpecialFunctions.Log.Base`, `Mathlib.MeasureTheory.Integral.Bochner.Set`

Setting. A measure space (X, μ) (`MeasureTheory.Measure`). The predicate `IsDensity` says that $\rho\colon X \to \mathbb{R}$ is measurable, pointwise nonnegative, integrable with $\int \rho\, d\mu = 1$, and that $\rho \log \rho$ is integrable. This is the space P_μ; as a type it is `Density`. The entropy S and the Jensen–Shannon divergence are in bits, and $d_1(\rho, \sigma) = \int |\rho - \sigma|\, d\mu$.

Results.

Brief	Lean
Lemma 4: $u^2/(2\ln 2) \le f(u) \le \lvert u\rvert$ on $[-1,1]$	`kernel_bounds`; via $f(u) = 1 - H_2(\frac{1+u}{2})$ (`kernel_eq`)
eqs. (15)–(17), pointwise	`jsdDensity_eq`, `jsdDensity_bounds`
$(\rho+\sigma)/2 \in P_\mu$ (used implicitly)	`isDensity_mid`
eq. (16): JSD as one integral	`jsd_eq_integral`
eqs. (18)–(19)	`l1dist_sq_le`: $d_1^2 \le 2\int(\rho-\sigma)^2/(\rho+\sigma)\,d\mu$
Prop. 10, eq. (11)	`jsd_bounds`
Prop. 10, eq. (12)	`tendsto_jsd_iff`, for any filter
eq. (26): $S(u_U) = \log_2 \mu(U)$	`entropy_unif`
eqs. (32)–(34): entropy of $(1-\varepsilon)u_B + \varepsilon u_A$	`entropy_mix`
eq. (31): $d_1(\rho_n, \rho) = 2\varepsilon_n$	`l1dist_mix`
Theorem 21	`entropy_not_l1_continuous`: $\rho_n \to u_B$ in L^1 and $S(\rho_n) \to S(u_B) + 1$; `not_tendsto_entropy`
second remark, eqs. (40)–(45)	`tendsto_entropy_mix_atTop`: $S(\rho_n) \to \infty$
Theorem 46	`eventually_jsd_lt` (JSD balls are τ-neighbourhoods), `eventually_l1dist_lt` (τ is finer than the L^1 topology)

Differences from the brief.

- Densities are functions, not L^1 classes, and are required to be measurable and nonnegative everywhere rather than almost everywhere. "$S(\rho) \in \mathbb{R}$" is read as integrability of $\rho\log\rho$.
- Lemma 4 is proved through the binary entropy. With $p = (1+u)/2$ it says $(2p-1)^2/2 \le \ln 2 - H(p) \le \ln 2\,|2p-1|$. The lower bound comes from concavity of $H(p) + 2(p-\frac{1}{2})^2$, using $H'' = -1/(p(1-p)) \le -4$, which is the brief's second-derivative argument. The upper bound comes from concavity of H, the brief's chord argument.
- Eq. (18) uses Cauchy–Schwarz. The formal proof uses the pointwise inequality $|a-b| \le \frac{(a-b)^2}{(a+b)\,2c} + \frac{c(a+b)}{2}$ with $c = d_1/2$, which gives the same bound $d_1^2 \le 2\int(\rho-\sigma)^2/(\rho+\sigma)$.
- Eq. (12) and the topological statements are formalized through convergence and neighbourhoods. The "topology generated by JSD balls" is not constructed; JSD is not a metric. The total-variation identity (2) is not formalized.
- In Theorem 21 the hypothesis $1 < \mu(A_n)$ is dropped. Only $\mu(A_n) \to \infty$ is used, and all statements hold eventually in n. The first remark (counting measure) is not formalized separately; it is the special case X countable.
- Theorem 46 assumes only that the *equal* mixture $(\rho,\sigma) \mapsto (\rho+\sigma)/2$ is continuous. That is weaker than the brief's hypothesis that all mixing is continuous, so the formal statement is stronger.

Problems found in the brief. No mathematical errors. Gaps the formalization had to fill:

- Prop. 10 and Theorem 46 use $(\rho+\sigma)/2 \in P_\mu$, i.e. that the mixture has finite entropy. The brief does not prove this. It follows from the pointwise bounds $0 \le$ JSD-integrand $\le \frac{\ln 2}{2}|\rho-\sigma|$ (`isDensity_mid`). Theorem 46 likewise presupposes that P_μ is closed under mixing.
- In Theorem 21, $\varepsilon_n = 1/\log_2 \mu(A_n) \in (0,1)$ needs $\mu(A_n) > 2$, not $\mu(A_n) > 1$. The brief's "passing to a tail" covers this.